

Development and Evaluation of an AR News Interface for Efficient Information Access

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Figure 1: News articles are displayed around the viewer to give an impression of being immersed in a wealth of information.

ABSTRACT

In this study, we developed a user interface (UI) for augmented reality (AR) glasses designed to allow users to browse news articles easily and efficiently. The UI uses natural language processing and dimensionality reduction techniques to place articles optimally within a virtual space. We compared this UI with two other AR-based interfaces in a user study with 13 participants, and the results show that the proposed interface reduced the time required to browse articles as well as the cognitive load of the activity.

Index Terms: Augmented Reality—News Interface—Information Retrieval—User Experience; Natural Language Processing—Dimensionality Reduction—

1 INTRODUCTION

As augmented reality (AR) technology becomes more prevalent in consumer markets, new interfaces that enhance everyday experiences like reading the news are increasingly needed. To overcome the limitations of traditional interfaces, we introduce a novel approach to allow users to browse news articles using AR glasses. The proposed method builds on foundational research in data visualization, AR interactions, and natural language processing, focusing in particular on providing news articles to the user with an efficient spatial arrangement and intuitive placement.

2 SYSTEM DESCRIPTION

We developed a new system to enable users to read the news efficiently in daily life using AR glasses with minimal effort. The proposed system is designed to meet emerging needs to lower the cognitive burden on users and extend the duration and frequency

of their engagement with relevant content while providing more efficient access to information from a wide variety of sources.

To this end, we developed a user-friendly interface for AR glasses to show an array of icons for news articles. Related news items are positioned close to each other in an immersive display, and chronological information is clearly visible. The system organizes news articles around the user as shown in Figure 1.

The positions of the displayed articles are selected as follows. To automate the process of arranging the news articles, we use the term frequency-inverse document frequency (TF-IDF) and perform dimensionality reduction via t-distributed stochastic neighbor embedding (t-SNE) [2]. TF-IDF quantifies each word in each article based on its frequency within the article and uniqueness across all articles. Thus, we obtain vectors in a high-dimensional space in which related articles are neighbors to vectorize the information comprising the articles. To display N news articles, let the TF-IDF vector of the i -th latest article be a_i , and let the corresponding coordinates in the physical space where icons are displayed be $b_i = (x_i, y_i, z_i)$. Here, x and z are horizontal relative to the ground and y is vertical, with the origin at the head of the person wearing the AR glasses. First, the system maps a_i onto a 2D horizontal plane.

We adopt t-SNE to perform nonlinear dimensionality reduction because TF-IDF vectors are nonlinear. In t-SNE, the similarity between a_i and a_j is modeled as the conditional probability that a_j is selected as a neighbor when a_i is given. Optimization is performed to match the conditional probability distribution as closely as possible with b_i and b_j . Thus, coordinates in reduced dimensions are obtained while preserving neighborhood relationships in a high-dimensional space. These 2D coordinates are then adjusted to x_i and z_i to fit within the display space.

The height y_i of the icons is determined using chronological information. Considering the position of the user's head as the origin and the height of the center of a display area as h , y_i is set to $y_i = -h(i-1)/(N-1)$. Thus, as i increases (i.e., as the news articles get older) the display moves downward to position the latest information closer to eye level. This setup allows users to browse the news efficiently.

3 USER STUDY

To evaluate the utility of the system, we prepared two additional display methods for comparison and interviewed a set of test users. The system was implemented on the Meta Quest Pro device. We asked users to perform recognition tasks for each display method to evaluate the efficiency and cognitive load of the system. Through this evaluation, we investigated the impact of displaying multiple pieces of information simultaneously and the effects of summarizing information using spatial arrangement.

3.1 Experimental Design

Here, we envisioned a scenario in which users routinely browse news articles with AR glasses. The experiment was conducted in setting resembling an ordinary living room, and the participants were seated. The participants could move the entire interface to their preferred

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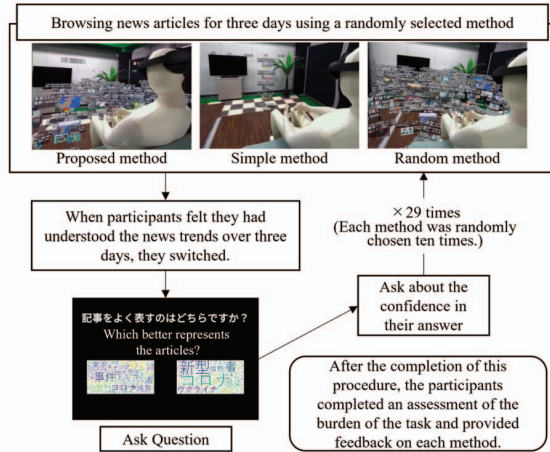


Figure 2: Schematic of the procedure.

positions using a controller. For a fair comparison, we asked the participants to try three different user interfaces.

Proposed Method In our proposed method, all news articles from a designated period are arranged in clusters of related articles. We also displayed pre-assigned genre tags above the median position of the articles in each genre. The proposed approach allows a user to simultaneously view multiple articles organized into meaningful groups. Participants were asked to navigate through a selection of news articles using a controller and select articles of interest for closer examination.

Simple Method This method was a basic user interface similar to conventional newsfeeds. The articles were displayed vertically by genre using the tags. Participants manipulated the articles by using a controller to toggle between genres by moving left and right and scroll through past articles by moving up and down. The number of articles that could be viewed simultaneously was limited by the available space, which ranged from eye level to the floor. Based on the settings, the number of articles displayed ranged from four to six.

Random Method To contrast the effects of the number of articles displayed with those of spatial summarization, we also experimented with a random method in which articles were arranged randomly in the virtual space. All the available articles were displayed simultaneously to take advantage of the wide display space of AR glasses. In contrast to the proposed method, x_i and z_i were set randomly. A random point was selected to obtain x_i and z_i within a circle with a radius equal to the width of the display space by using the Random-insideUnitCircle function in the Unity framework. Participants navigated by using a controller to select articles.

3.2 Task

To assess the activity of browsing multiple news articles, we formulated a task to measure whether participants could recognize the overall trends in a set of news articles accurately. We used word clouds to represent the trends and asked participants to select the word cloud representing the information they viewed. Word clouds and word tags are widely used visualization techniques for textual information. In this step, we used the Python WordCloud package [1].

Thirteen participants (11 men and 2 women, with an average age = 33.1 ± 6.2 (SD)) were recruited for the experiment. The news articles used were sourced from NHK News Web and had been published between September 30, 2022, and April 11, 2023. The

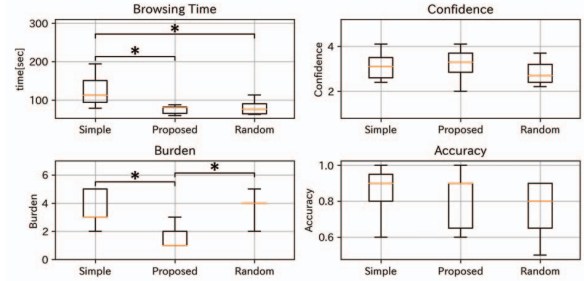


Figure 3: Results of each method. An asterisk indicates that $p < 0.05$.

flow of the task performed by the participants is shown in Figure 2. The average number of articles displayed was 282 ± 51 (SD).

4 RESULTS

We analyzed data from 11 of the 13 participants; one of the original 13 withdrew due to health concerns and data from another was excluded due to their lack of engagement with the activity. We found no significant differences in terms of accuracy and confidence levels across the three interfaces, as confirmed by a one-way ANOVA at a 5% significance level. However, we did find significant differences in terms of browsing time and user burden. Longer browsing times were recorded for the "Simple" method compared to the "Proposed" and "Random" methods, with significant differences confirmed by one-way ANOVA and Tukey-Kramer post-hoc tests. The proposed method showed a lower user burden compared to "Simple," which was also confirmed as significant using the same statistical methods.

5 DISCUSSION

While accuracy and confidence levels were comparable across the three methods, significant differences were observed in terms of browsing time and user burden. The proposed method was the most efficient owing to the spatial clustering of articles, which allowed for quick identification and access to relevant news. The simple method was found to be more time-consuming and burdensome, which highlights the limitations of conventional linear displays for newsfeeds in AR settings. Interestingly, the random method offered some unexpected benefits in discovering diverse content despite its lack of structured organization, which suggests that incorporating serendipity into news browsing experiences could be useful. These insights highlight the complex interplay between information arrangement, user cognitive load, and the overall experience of consuming news content in AR environments. The proposed method was found to be the most usable because it displayed a wide array of content across a 3D virtual space while facilitating easier browsing. Our results show that this approach provides an example of an effective user interface for AR news applications.

6 CONCLUSIONS AND FUTURE WORK

The developed UI demonstrates the potential for efficiently browsing news articles using AR glasses with a low cognitive burden. In future research, we plan to compare this interface with smartphone and PC solutions and consider optimal parameters to display a wide variety of information with an intuitive and simple user interface in AR environments.

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